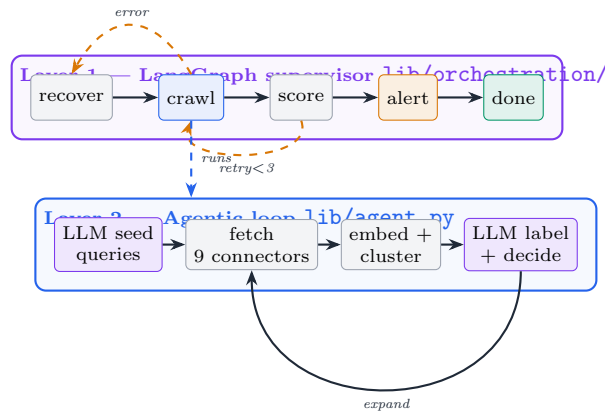


# PulseTrace — Agentic Multi-Source Sentiment Intelligence

Topic in  $\rightarrow$  LLM agent loops  $fetch \rightarrow embed \rightarrow cluster \rightarrow label$ , decides itself when coverage has **converged**  $\rightarrow$  ranked evidence, sentiment timeline & cited RAG Q&A, inside a fault-tolerant LangGraph supervisor.

## 1 ONE PICTURE — a resilient shell around an adaptive loop



**Say this:** “It’s *two layers*. LangGraph isn’t the search brain — it’s the fault-tolerant supervisor (retry / alert / schedule). The agentic search lives inside one node, **crawl**. Separating ops from intelligence makes a failure *route* to **recover** instead of crashing.”

## 2 THE WOW — the LLM is in the loop *twice*, and stopping is *measured*

**Seed.** One LLM call  $\rightarrow$  5 diverse queries (6 in opinion mode: half support, half challenge).

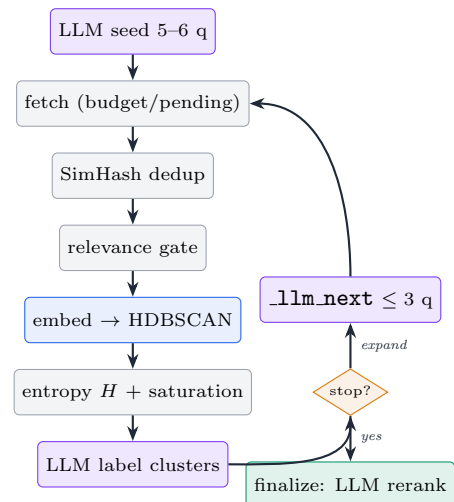
**Fetch.** All pending (**query, source**) in a thread pool; per-query limit shrinks so a run never blows the 500-post budget.

**Dedup.** 64-bit SimHash, Hamming  $\leq 10$  — reposts don’t burn tokens or double-count clusters.

**Gate.** Drop posts with topic-overlap  $< 0.12$  — *but only if  $\geq 6$  survive* (recall floor).

**Cluster.** Gemini embeds  $\rightarrow$  HDBSCAN (finds  $k$  itself); KMeans fallback.

**Decide.** Labels (not raw posts)  $\rightarrow$  **llm\_next** proposes  $\leq 3$  queries, each re-anchored to the core subject so it can’t drift.



**Stop if ANY:** budget  $|seen| \geq 500$  • iters  $\geq 4$  • *converged*  $|H - H_{-1}| < 0.05$  • *saturated*  $sat \geq 0.80$  • LLM says stop. **Say this:** “Thin topic converges in 2 iterations; a rich topic expands to the budget. *The agent decides how hard to look* — it’s not a fixed page count.”

## 3 THE PROOF — relevance-dominant ranking, measured, not guessed

$$\text{score} = 0.85 \text{ rel} + 0.08 \text{ eng} + 0.05 \text{ rec} + 0.02 \text{ src} \\ \times 0.3 \quad \text{if rel} < 0.35$$

Two stages: a *cheap pure rank.posts* runs every iteration (keeps the live UI top-posts fresh, builds a 30-shortlist), then *one llm.rerank* scores 0–100 with a prompt-injection guard. Sub-0.35 is buried  $\times 0.3$ , not deleted.

**Say this:** “We *measured* that engagement was noise against the judge, re-weighted relevance  $0.60 \rightarrow 0.85$ , and beat the #1 GitHub agentic engine  $0.716$  vs  $0.520$  on identical model/sources/judge. The edge is the loop, not the model.”

## 4 WHY NOTHING SINGLE-POINTS-OF-FAILURE & THE NUMBERS

**Resilience chain (every layer degrades, none crashes):**

- connector fails  $\rightarrow$  log & skip
- HDBSCAN can’t cluster  $\rightarrow$  KMeans
- Gemini key 429 / spend-cap  $\rightarrow$  backup key, sticks
- pipeline throws  $\rightarrow$  graph routes to **recover** (retry $\times 3$ , 30s backoff)

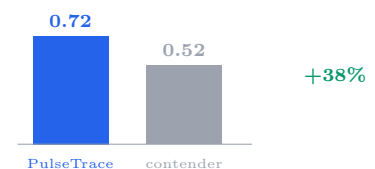
**9 connectors** behind one ABC (reddit, hn, fb, x, ig, youtube, polymarket, github, bluesky). Reddit+HN always-on; FB needs cookies; X/IG return [] until keyed.

**Measured profile** (prod, 500-post run):

|                            |                              |
|----------------------------|------------------------------|
| fetch $\times 3$           | $\sim 11s$                   |
| embed + cluster $\times 3$ | $\sim 18s$                   |
| label $\times 3$           | $\sim 9s$                    |
| final LLM rerank           | $\sim 12s$                   |
| briefing + evidence        | $\sim 9s$                    |
| <b>total end-to-end</b>    | <b><math>\sim 62s</math></b> |

$\sim 9k$  LOC production Python • 386 tests • 15 MCP tools • live HTTPS.

nDCG@5 vs the #1 OSS agent



## 5 30-SECOND CLOSE — the five lines

**1. Two layers.** LangGraph = resilience; **run\_agent** = intelligence. Don't conflate. **2. LLM in the loop twice.** Seeds queries, then re-plans from the clusters it just found — that's agentic, not a scrape. **3. Stopping is measured.** Entropy + saturation decide convergence. **4. Ranking is relevance-dominant by evidence.** +38% over the best OSS engine, same conditions. **5. Nothing single-points.** Connector → skip; HDBSCAN → KMeans; key → backup; throw → recover.